

* Activity - 1 *

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Q. ~~Push(54) & Pop(52)~~

1.

push(54) $\rightarrow$ [54]

push(52) $\rightarrow$ [54, 52]

pop() $\rightarrow$ [54]

push(55) $\rightarrow$ [54, 55]

push(62) $\rightarrow$ [54, 55, 62]

s = pop() $\rightarrow$ [54, 55]

$\therefore$ s = 62

enqueue[21] $\rightarrow$ [21]

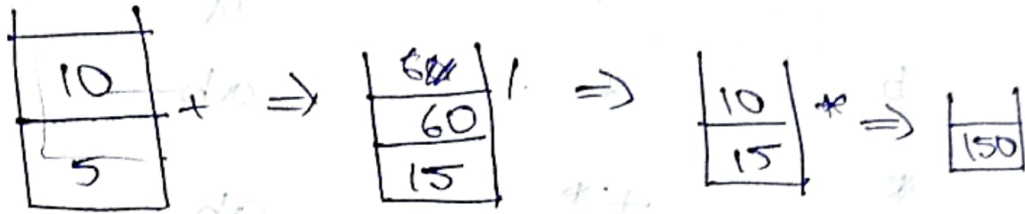
enqueue[24] $\rightarrow$ [24, 24]

q = dequeue() $\rightarrow$ [24]

$$S + a_j = 62 + 22 = 84$$

$$= 9086$$

2. $105 + 606 / * 8 -$



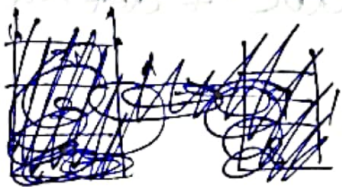
$$10 + 5 \rightarrow 15$$

$$60 \cdot 15 \rightarrow 900$$

$$15 \times 10 \rightarrow 150$$

$$150 - 2 \rightarrow 148$$

3. $223 \wedge / 23 * + 51 * -$



Push (8)

Push (2)

push (3)



$\Rightarrow \begin{bmatrix} 8 \\ 8 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 6 \\ 1 \end{bmatrix}$

$\Rightarrow \text{Pop} (1), \text{Pop} (2) \Rightarrow \begin{bmatrix} 6 \end{bmatrix}$

$\Rightarrow 61$

$[0] \leftarrow [0] \text{ New}$

$[0] \leftarrow [0] \text{ New}$

$[0] \leftarrow [0] \text{ New}$

$[0] \leftarrow [0] \text{ New}$

4.

$$a + b \times c - d \wedge e \wedge f$$

2800:

char.

stack

output

a

+

a

a

b

+

ab

*

+

ab

c

+

abc

-

-

abc*

d

-

abc*+d

^

-

abc*+d^

e

-

abc*+de

f

-

abc*+def

empt

end

abc*+def^

Postfix eq: $abc^* + def^{\wedge} -$

5.

push(10) $\rightarrow$ [10]

push(20) $\rightarrow$ [10, 20]

pop() $\rightarrow$ [10]

push(10) $\rightarrow$ [10, 10]

push(20) $\rightarrow$ [10, 10, 20]

pop() $\rightarrow$ [10, 10]

pop() $\rightarrow$ [10]

pop() $\rightarrow$ []

push(20) $\rightarrow$ [20]

pop() $\rightarrow$ [20]

Popped value = 20, 20, 10, ~~8~~¹⁰, 20

Amf
22/7